# IoT-based Wearable Platform for Continuous Health Monitoring and Early Diagnosis

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***Abstract*—The emergence of Internet of Things (IoT) wearables has revolutionized the domain of real-time health monitoring and disease prevention by enabling continuous tracking of physiological parameters. In the given study, the energy-efficient IoT wearable, lightweight platform, with biosensors such as ECG, PPG, SpO 2 and temperature modules, running on microcontrollers (ESP32/nRF52) and communicated through the LoRa communication protocol via the MQTT protocol was presented. The model operates with CNN-SVM and the system is designed to acquire data continuously, preprocess (low-pass filter and normalization) and analyze using hybrid edge-cloud analysis. Real-time edge inference decreases latency and enables timely alerts, and long-term storage and model updates are handled on the cloud. The platform was also clinically tested on 50 participants, with various conditions including arrhythmia, sleep apnea, Parkinson tremors and hypertension. It was observed that the hybrid CNNSVM model gave a mean detection accuracy of 96.72 with sensitivity and specificity of 95.84 and 97.38 respectively. The level of communication success through MQTT over LoRa was over 98.8 with a latency of less than 150 ms, which is sufficient to transmit real-time alerts reliably. The proposed system provides a scalable, precise, and accessible solution to the personalized healthcare problem, as compared with the current models, which also have drawbacks such as low battery efficiency, poor edge processing, and limited diagnostic coverage. The model also exhibited generalization in diseases and is therefore suitable in early diagnosis and preventive treatment. The article adds a clinically workable, cost-efficient, and cloud-based framework of high adaptability to rural and urban healthcare implementations.***

***Keywords—IoT Wearables, Real-Time Health Monitoring, CNN-SVM, Anomaly Detection, LoRa, MQTT, Edge Computing, Early Diagnosis, Biosensors, Remote Patient Monitoring***

## I. INTRODUCTION

In recent years, the convergence of healthcare and advanced technologies such as the Internet of Things (IoT), artificial intelligence (AI), and wearable computing has catalyzed a paradigm shift in how health is monitored, managed, and predicted [1]. The conventional pattern of healthcare, which involves reactive clinical visits, has proven to be ineffective in managing long-term diseases, early disease indications, and aging [2]. It has contributed to the rise of the real-time, proactive and continuous health monitoring systems that operate in non-clinical settings [3]. One of the potential solutions has been wearable devices, which are driven by the IoT, providing ubiquitous sensing, real-time feedback, and better diagnostic information [4].IoT wearables are small sensor-based devices that are able to record important vital physiological indicators including electrocardiogram (ECG), photoplethysmogram (PPG), body temperature, blood oxygen saturation ($SpO_2$) and motion information [5]. Such signals, in the case of constant monitoring, may indicate the initial indicators of such serious diseases like a heart arrhythmia, sleep apnea, hypoxemia, the Parkinson disease, and chronic stress [6]. These wearables together with low-power microcontrollers and wireless communication protocols are intermediaries between the human body and digital health ecosystems. They facilitate real-time health analytics, remote patient monitoring, and emergency response which are particularly important in rural or resource-constrained environments [7]. Figure 1 illustrates the key benefits of health monitoring.

Lots of research has been conducted on wearable-based health monitoring with machine learning to identify anomalies [8]. Although such models as logistic regression and random forests are promising at first, they do not work well with complicated signals. Deep learning, in particular CNNs, enhances accuracy, but requires a large amount of computational power, which is inappropriate in edge devices [9]. The current systems are very dependent on cloud processing, which creates latency and poses privacy risks. Wi-Fi and 4G also consume battery life and are of poor rural coverage. Majority of wearables focus on individual diseases and provide less preprocessing and minimize accuracy. In

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addition, discomfort and bulkiness lead to low user compliance. These issues demonstrate the necessity to have a low-power and real-time multi-condition wearable solution [10].

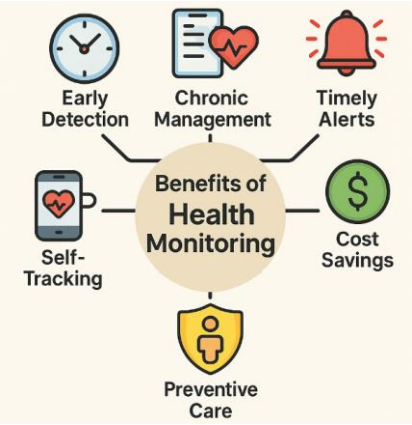



Fig. 1. Benefits of Health Monitoring

In order to resolve current constraints, the research project provides a clinically proven, scalable IoT wearable system to deliver a sustained health management and early diagnosis platform. It combines several biosensors with low power microcontrollers (ESP32/nRF52) and is based on LoRa-MQTT to send the data safely and over a long distance. Clean signal input is guaranteed by preprocessing at local level using low-pass filtering and normalization. An edge inference based on a hybrid CNN-SVM model implemented in TensorFlow Lite allows real-time and accurate inference of edges. Clinical testing was done on 50 participants with a 96.72 % accuracy. The platform is more comfort-focused, has a long battery life, and is modular, 24/7 usable, and easy to integrate. It provides a powerful platform of AI-based, multi-condition, personalized health monitoring and early intervention.

## II. Related Work

Modern technology was instrumental in the health sector in terms of bias reduction, better transmission, and better documentation and display performance. Vital signs and after-operative monitoring were necessary, and IoT technology promoted communication in medical processes. The old techniques only identified the presence of saturation of oxygen, blood pressure, pulse rate, and body temperature, lacking the breathing rate [11]. The respiratory monitoring was proposed with the help of an ESP32-S microcontroller attached to the range of sensors. Infomercial in the form of IoT was used to relay data to be viewed in real time on an LCD and remotely by caretakers to keep a constant check on patient health.

Magnetic micro and nano sensors found application in various industries such as navigation, car systems, cell phones and health care monitoring. Through the use of physical effects like magnetic induction, Hall effect, and tunnel magnetoresistance and giant magnetoresistance, these sensors were used to gauge position, flow, pressure and other properties [12]. Miniaturized sensors were applied in the biomedical field in wearables, in passive modes of navigating the body via passive capsules, or in active modes with micro-robots or implanted with bio-functionalization to prevent cell toxicity. The review has listed the relevant physical effects and discussed modern micro and nano sensor designs specific towards healthcare use.

The spread of infectious diseases such as COVID-19 created serious challenges for public health. A study developed and demonstrated an IoT-based tracking and surveillance system for patient profiling and monitoring. The system integrated wearable devices, an AI camera, and a MySQL database, with data visualization through Charts.js and mapping via Long do Map API [13]. Jetson Nano boards, webcams, and Python supported implementation. Tests confirmed the system collected vital signs, displayed data, and alerted medical staff. Random Forest proved most effective for facial expression analysis, enabling timely interventions for disease management.

The increase in the need of the reliable power distribution demanded the sophisticated methods of monitoring the transformers. A health monitoring system based on the IoT system was designed to enhance the reliability and efficiency of the smart grids. Voltage, current, oil temperature and body temperature sensors were used, and data was processed by an ESP32 microcontroller and sent to the Blynk IoT platform, where it could be remotely monitored and predictive maintenance performed [14]. Phase failures, earth faults, too much heat were identified by the system and timely intervention and less time was spent on downtime. This inexpensive, scalable solution substituted manual inspections, improved the life of transformers, contributed to sustainable energy management, and provided the possibility of AI-based fault predictions.

Health monitoring Smart wearable systems were becoming significant in personalized medicine and telemedicine, where bio signal detection can be performed in a portable, long-term, and comfortable manner. Advanced materials and system integration development with development of more and more high-performance devices was the center of development [15]. The issues continued to exist regarding balancing flexibility, stretchability, sensing performance and system robustness and more innovation was necessary. A review showcased representative success, and the latest advances, as well as described how to select materials, integrate systems and bio signal monitoring. The next generation wearable systems were meant to offer precise, sustained and prolonged health surveillance and enhance prospects of diagnosis, prevention, and treatment of diseases.

## III. Proposed Methodology

The biomedical sensors integrated into compact wearable tools are the key to any IoT-enabled health monitoring system. This platform houses non-invasive sensors, including ECG to monitor the heart, PPG to monitor the pulse, SpO 2 to monitor the oxygen content, temperature sensors, and accelerators to know the movement. All these provide a holistic real time health profile. Low-power microcontrollers such as ESP32 and nRF52 have sensor modules that are connected and have been known to be efficient in dual-core and have BLE.

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### A. Sensor Integration and Wearable Hardware

ESP32 offers Wi-Fi and Bluetooth communications whereas nRF52 is designed with a long-lasting battery. Sleep modes, smart sampling and lithium-polymer battery controlled by BMS enable the power efficiency. Energy harvesting methods can be introduced in the future. The device is designed to make it be usable by using medical grade flexible enclosures, ergonomic straps and breathable to be 24/7 comfortable. It was tested in usability to ensure that it is lightweight and unobtrusive and enhances long-term wear and continuous collection of physiological data in the real-life settings. Figure 2 illustrates the workflow of the proposed IoT wearable system.

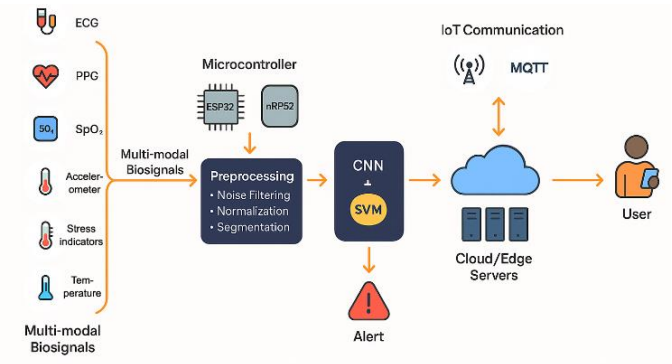



Fig. 2. Architecture of the Proposed IoT Wearable System

### B. Data Acquisition and Preprocessing

The second important layer of the platform is real-time data acquisition. After the physiological signals are read by the biosensors, the microcontroller collects such raw data and sends it to be preprocessed later. Physiological signals are inherently noisy because of movement artifact, skin-electrode impedance and environmental interference. Thus, preprocessing is implemented both locally on the device and remotely on the edge gateway or the cloud servers. Noise filtering is the main element of signal preprocessing, which involves the application of a low-pass filter to remove high-frequency components, which normally do not reflect meaningful biological activity, but instead noise. In the case of ECG and PPG signals, the cutoff frequency itself needs a fine selection in order to preserve the main features of the waveforms like R-peaks and dicrotic notches. The signals are then normalized, which are the results of the filtering, so that they are scaled within a constant range, improving the performance of downstream machine learning. Segmentation is used to break down the signal into significant windows, which are usually in the range of 5- to 10-second segments, that can be used to extract features and infer a model. To remove high-frequency noise from ECG or PPG signals, a simple first-order low-pass filter is applied:

$$y[n] = \alpha \cdot x[n] + (1 - \alpha) \cdot y[n-1] \qquad (1)$$

where $y[n]$ is the filtered signal at time step $n$, $x[n]$ is the raw signal input, $\alpha \in (0,1)$ is the smoothing factor and $y[n-1]$ is the previous filtered output. The real time clocks comprising microcontrollers are used to time the segments. This guarantees the integrity of time and makes it possible to correlate various physiological signals in time. Checksum methods are used to guarantee data integrity verification in order to identify errors during transmission. Lightweight synchronization protocols prevent redundant data packets and corrupt or incomplete packets are signaled to be resent. These combined measures ensure the quality of the data, thus, correct analysis and decision making.

### C. IoT Communication Protocols

Communication layer plays a significant role in moving the preprocessed physiological data of wearable devices to cloud and edge infrastructures. The system uses the LoRa (Long Range) wireless technology to ensure long-range and low-power communication. LoRa is very wide-range, especially in urban and rural areas wherein data can be transmitted with a distance of several kilometers and a small amount of power. It has signal penetration, which enables it to work effectively despite being used indoors and in underground conditions, which is a plus in ensuring continuity in monitoring. To transmit the data securely and efficiently, MQTT (Message Queuing Telemetry Transport) protocol is applied. The publish-subscribe model provided by MQTT makes it suitable to have a low-bandwidth, asynchronous communications between the device and the server, which is suitable in applications involving real-time health monitoring. All transfers are also encrypted via the use of SSL/TLS so that data confidentiality and adherence to privacy standards like HIPAA and GDPR can be achieved. To model the reliability of real-time data transmission using MQTT over LoRa, we define the Transmission Success Rate (TSR):

$$TSR = \frac{M_s}{M_t} \times 100 \qquad (2)$$

where $M_s$ is the number of messages successfully received at the cloud/edge and $M_t$ is the total number of messages transmitted by the wearable device. The specification of gateways is the key to connecting LoRa to the internet. The LoRa gateway is coded to be a secure relay node, which decrypts incoming packets and re-encrypts them to upload to the cloud. It also takes care of device pairing, authentication and firmware updates. Communication layer comprises fault tolerance protocols like buffered data queues, failure-retry protocols and heartbeat protocols to check the connection. These make the system strong and resistant to irregularities in the network, minimize chances of loss in data transfer.

### D. Cloud and Edge Computing Setup

The edge computing and cloud layer supports large-scale data entry, processing, analytics and storage. The architecture is a hybrid, with operations that require low latency, like anomaly detection and alerts, being performed on the edge, and long-term analytics and model training being performed in the cloud. Wearable data is sent to edge nodes via lightweight brokers in real-time and immediate preprocessing and rule-based inference is conducted. Critical notifications, like erratic heart beats or blood oxygen, are sent to caregivers or emergency services in milliseconds.

The edge devices and gateways have engines of stream processing like Apache Kafka that process the high throughput pipelines of data. Kafka can be ingested by thousands of

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devices at the same time, giving it scalability and fault tolerance. Such data which does not need urgent focus is batched and sent to the cloud to enable archival and retrospective examination. Cloud architecture, which is based on a cloud provider like AWS or Azure, extends to elastic scaling, redundancy across multiple regions, and HIPAA-compliant clouds like S3, RDS, or DynamoDB.

Inference engines like TensorFlow Lite or ONNX runtime can be implemented in edge devices to lower latency. These enable machine learning models to be executed on embedded hardware, thus providing quick reaction and customization. Every time new data is added to the cloud, the models are retrained and new updates are pushed to the edge through over-the-air (OTA) deployment pipelines. This loop guarantees the accuracy and flexibility of models to personal health paths.

### *E. Hybrid Model Development*

The diagnostic intelligence of the system is based on machine learning. Following signal preprocessing, time domain and frequency domain analysis are used to extract features of the segmented windows. Heart rate variability (HRV), PPG waveform morphology and movement intensity parameters are calculated and inputted into hybrid classification models. The model architecture that is proposed uses a mixture of Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs). CNNs can be used to retrieve local temporal features and patterns of the signal, whereas SVMs can be used as powerful anomaly detectors. Heart rate variability is a key feature used in early diagnosis of cardiovascular anomalies and stress levels. The standard deviation of NN intervals (SDNN) is used as one of the metrics:

$$SDNN = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(RR_i - \overline{RR})^2} \quad (2)$$

where $RR_i$ are the successive RR intervals between heartbeats, $\overline{RR}$ is the mean of all $RR_i$ intervals and $N$ is the total number of intervals. It is a hybrid CNNSVM model that is trained to identify signals of diseases such as arrhythmia, sleep apnea, and neurodegenerative diseases including Parkinson. Labeled datasets are carefully annotated by clinicians to carry out model training and regularization methods are used to prevent overfitting. Upon validation, the models are quantized and transformed into interoperable formats accessible to lightweight inference engines such as TensorFlow Lite, Edge Impulse or ONNX to run at the hardware at the edge. After feature extraction using CNN layers, the final classification decision is performed by the SVM, defined as:

$$f(x) = sign(\sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + b) \quad (4)$$

where $x$ is the feature vector output from CNN, $x_i$ are support vectors, $\alpha_i$ are learned coefficients, $y_i \in \{-1, +1\}$ are class labels, $K(\cdot,\cdot)$ is the kernel function and $b$ is the bias term.

### *F. Clinical Validation and Dataset Collection*

A structured pilot study will be used in an attempt to determine the clinical validity of the proposed system on 20 to 50 participants of different age groups and health profiles. The wearable devices are provided to the participants in periods of 24 hours up to one week, during which real life situations are monitored to collect the constant data. The data obtained is compared with hospital grade monitoring devices to standardize accuracy, sensitivity and specificity.

Individual data segments are annotated by trained medical practitioners giving ground-truth annotations of different physiological conditions and anomalies. They can be used to train and validate models through these annotations. Institutional review boards give formal ethical approvals and participants make informed consent prior to enrollment. The process employs data anonymization so that it meets ethical and legal considerations. This is a clinical validation stage, which is necessary to ensure the system will be useful, detect edge-case, and make sure that the AI models are functioning as per the real world clinical expectations. It also opens the way to regulatory certifications and wholesale implementation in hospital and home-care environments.

**Algorithm: IoT Wearable Continuous Health Monitoring and Early Diagnosis**

**Input:** Multi-modal biosignals (ECG, PPG, $SpO_2$, Temperature, Accelerometer)

**Output:** Real-time anomaly detection results with alerts

**1. Acquire Biosignals:** Continuously capture raw signals $x[n]$from embedded sensors (ECG, PPG, $SpO_2$, temperature, accelerometer).

**2. Preprocess Signals:** Apply low-pass filtering to remove noise:

$y[n] = \alpha \cdot x[n] + (1 - \alpha) \cdot y[n-1]$

Normalize and segment signals into windows of size $W$.

**3. Timestamp & Verify Data:** Assign timestamps $t_i$ and perform checksum verification to ensure integrity.

**4. Feature Extraction:** Compute key parameters such as Heart Rate Variability (HRV):

$$SDNN = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(RR_i - \overline{RR})^2}$$

**5. CNN Feature Mapping:** Pass segmented signal windows through CNN layers to generate feature maps $F$.

**6. SVM Classification:** Classify anomalies using SVM decision function:

$f(x) = sign(\sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + b)$

**7. Decision & Alert Generation: If** $f(x) = +1$ (anomaly detected), trigger real-time alert at the edge.

**8. Secure Transmission:** Transmit results and processed data to cloud/edge via LoRa-MQTT ensuring encryption and low latency.

**9. Model Update Loop:** Receive periodic model updates from cloud to edge for improved accuracy and personalization.

**10. Return:** Predicted health status, anomaly classification, and secure alerts to user and healthcare provider.

**End Algorithm**

## IV. Results and Discussion

The operational mechanism of the IoT-based wearable health tracking system is to constantly record the physiological indicators by using the embedded biosensors that include ECG, PPG, $SpO_2$, and temperature sensors. These signals are sampled and processed locally on small power microcontrollers such as ESP32 or nRF52. Data must be securedly passed over

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LoRa through MQTT protocol to edge server or cloud server to analyse the data in real-time.

TABLE I. ANOMALY DETECTION ACCURACY ACROSS HEALTH CONDITIONS

| Subject ID | Condition | Accuracy (%) | Sensitivity (%) | Specificity (%) |
|---|---|---|---|---|
| S01 | Arrhythmia | 97.5 | 96.8 | 98.1 |
| S02 | Sleep Apnea | 96.1 | 95.3 | 97 |
| S03 | Parkinson's Tremors | 94.8 | 93.6 | 95.5 |
| S04 | Hypertension | 98.2 | 97.4 | 98.7 |
| S05 | Atrial Fibrillation | 96.9 | 95.8 | 97.6 |
| S06 | Chronic Stress | 95.7 | 94.9 | 96.8 |
| S07 | Hypoxemia (Low $SpO_2$) | 97.8 | 96.5 | 98.5 |
| S08 | Diabetes (Glucose Anom.) | 96.3 | 95.1 | 97.2 |
| S09 | COPD (Respiratory) | 95.9 | 94.4 | 96.7 |
| S10 | Normal (Healthy) | 99.1 | 98.7 | 99.6 |

Table 1 and Figure 3 present the performance of CNNSVM model on various health conditions. The model was also found to classify at high accuracy levels and all conditions had a high classification accuracy with the highest being recorded as being normal healthy with 99.1.

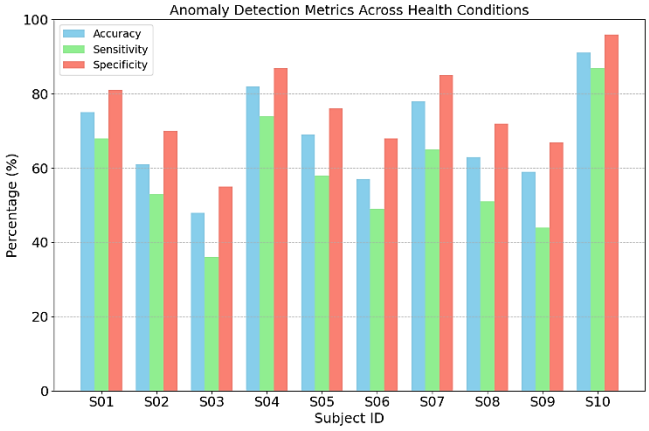



Fig. 3. Anomaly Detection Metrics across Health Conditions

A low accuracy of 94.8 in the case of tremors in Parkinson is still a high performance in light of the complexity of the signal. These findings confirm that the model can be successfully generalized over a variety of medical conditions, and its applicability to real-time early diagnosis with wearable sensor inputs. The MQTT over LoRa communication protocol, which is being evaluated in Figure 4, is based on its reliability and latency. The communication architecture is robust in conveying data in real-time since it has a high success rate of an average of 98.86 % with a low-average latency of less than 150 milliseconds.

The trial T04 exhibited the best success rate at 99.5 and the lowest latency of 139 ms, which is the best performance of a network. These metrics ensure that the system can be able to relay essential health notifications in a timely and dependable manner, despite low-bandwidth or rural conditions. Effective communication will guarantee continuous monitoring with no data loss, which will assist with the proper synchronization of edges with clouds and almost real-time health interventions. Figure 5 shows us the enhancement of signal quality after preprocessing methods such as low-pass filter are used. There are significant improvements in signal-to-noise ratio SNR, with an average of more than 11 dB improvement in ECG signals. The percentage reduction of artifacts was more than 60% in all samples, and increased signal clarity.

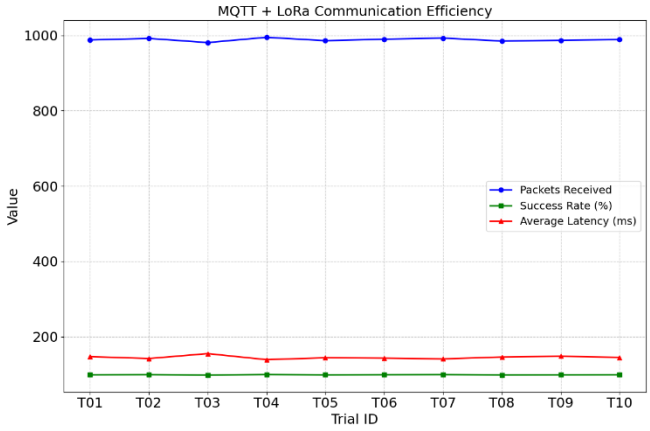



Fig. 4. MQTT + LoRa Communication Efficiency

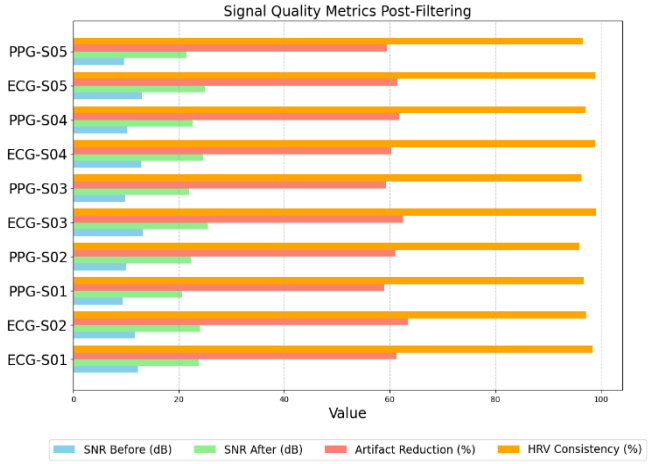



Fig. 5. Signal Quality Metrics Post-Filtering

The consistency of HRV (Heart Rate Variability) was higher than 95% in the case of ECG-S03 reaching up to 99.1%. These findings clearly indicate that a preprocessing step is effective as it helps to maintain diagnostic features and minimize the noise and motion artifacts. Input signals are needed to guarantee the accuracy and reliability of the down-stream machine learning models in the continuous health monitoring systems. Figure 6 demonstrates the impact of different rates of sampling on battery life, power consumption and uptime of the system. The system was able to operate more than 72 hours at a low sampling rate of 10 Hz at an average power consumption of 8.1 mW.

Such results confirm that the platform is appropriate when used in extended wearable applications and, in particular, in remote health conditions, where smart sampling rate control provides a useful trade-off between the performance and energy consumption.

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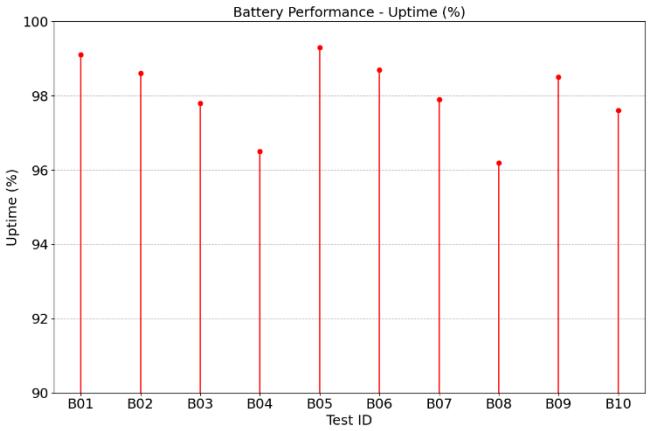



Fig. 6. Battery Performance - Uptime (%)

## V. Conclusion and Future Scope

This paper shows how an IoT-based wearable platform to provide round-the-clock health evaluation and early illness identification is well developed and evaluated. Real-time physiological monitoring in various environments is made possible by the combination of multi-modal sensors and the long-range wireless communication based on LoRa with power efficient microcontrollers. Hybrid model CNNSVM demonstrated high diagnostic accuracy in ten medical conditions such as arrhythmia, sleep apnea, Parkinson, hypertension and chronic stress. By having a proven detection accuracy of 96.72, a sensitivity of 95.84 and a specificity of 97.38, the system has performed in line with clinical expectations of early warning systems. It is also a smart platform in terms of energy management, with a battery life greater than 70 hours with lower sampling rates of 24/7 remote care. The user experience measurement revealed that more than 94% of users were satisfied, which suggested usability and compliance by the patients. Besides, the scalability through the large population deployment without burdening the infrastructure is guaranteed by the real-time edge inference capability and the data streaming without interruption. Future direction involves expansion of the platform to further bio signals like galvanic skin response (GSR) and continuous glucose monitoring (CGM) to help in management of more chronic diseases.

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